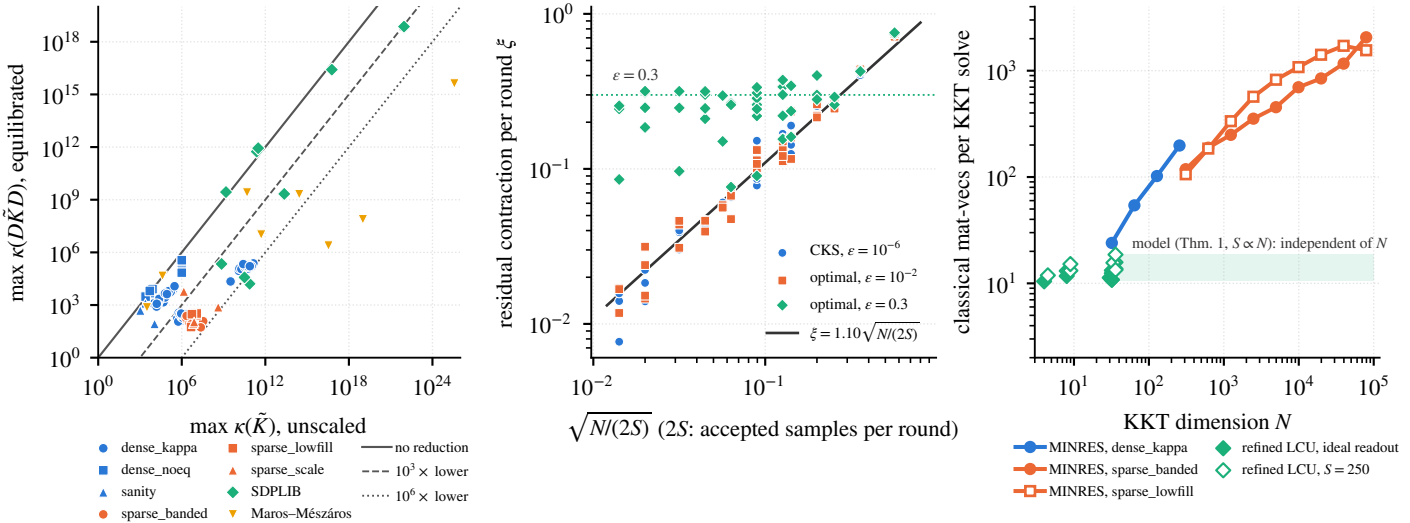
(a) effect of equilibration
max $\kappa(D\tilde{K}D)$, equilibrated
max $\kappa(\tilde{K})$, unscaled
dense_kappa
dense_noeq
sanity
sparse_banded
sparse_lowfill
sparse_scale
SDPLIB
Maros–Mészáros
no reduction
$10^3\times$ lower
$10^6\times$ lower
(b) readout model, 9 systems
residual contraction per round ξ
$\varepsilon = 0.3$
CKS, $\varepsilon = 10^{-6}$
optimal, $\varepsilon = 10^{-2}$
optimal, $\varepsilon = 0.3$
$\xi = 1.10\sqrt{N/(2S)}$
$\sqrt{N/(2S)}$ ($2S$: accepted samples per round)
(c) classical work per Newton solve
classical mat-vecs per KKT solve
model (Thm. 1, $S \propto N$): independent of N
KKT dimension N
MINRES, dense_kappa
MINRES, sparse_banded
MINRES, sparse_lowfill
refined LCU, ideal readout
refined LCU, $S = 250$